

Extremal Analysis of the Product-Sum Diophantine Equation

Rigorous Bounds and Realizability in $\mathbb{Z}^+$

Abstract

We establish the sharp upper bound on the maximum coordinate of an n -tuple of positive integers $(x_1, x_2, \dots, x_n)$ whose product equals its sum. Using reciprocal transformations, monotonic orderings, and product factorizations, we present a complete proof demonstrating that for all integers $n \geq 2$, $\max_{1 \leq i \leq n} x_i \leq n$, with equality achieved uniquely up to permutation by the configuration $(1, 1, \dots, 1, 2, n)$.

1 Problem Statement

Problem 1. Let $n \geq 2$ be an integer. Determine the maximum possible value of

$$\max\{x_1, x_2, \dots, x_n\}$$

over all tuples of positive integers $(x_1, x_2, \dots, x_n) \in (\mathbb{Z}^+)^n$ satisfying the Diophantine equation

$$\prod_{i=1}^n x_i = \sum_{i=1}^n x_i. \quad (1)$$

2 Main Result

Theorem 1. For any integer $n \geq 2$ and any solution $(x_1, x_2, \dots, x_n) \in (\mathbb{Z}^+)^n$ to Equation (1),

$$\max_{1 \leq i \leq n} x_i \leq n.$$

Furthermore, this upper bound is sharp, and the maximum value n is achieved by the multiset

$$\{x_1, x_2, \dots, x_n\} = \{1, 1, \dots, 1, 2, n\}.$$

3 Rigorous Derivation and Proof

Proof. Let $(x_1, x_2, \dots, x_n)$ be an arbitrary solution in positive integers to Equation (1).

Step 1: Permutation Invariance and Ordering

Because both the elementary symmetric product $e_n(x_1, \dots, x_n) = \prod_{i=1}^n x_i$ and the elementary symmetric sum $e_1(x_1, \dots, x_n) = \sum_{i=1}^n x_i$ are symmetric functions invariant under any permutation $\sigma \in S_n$, we may assume without loss of generality that the coordinates are sorted in non-decreasing order:

$$1 \leq x_1 \leq x_2 \leq \dots \leq x_{n-1} \leq x_n. \quad (2)$$

Under this convention, the quantity to maximize is precisely x_n .

Step 2: Reciprocal Transformation

Dividing both sides of Equation (1) by the strictly positive product $P = \prod_{j=1}^n x_j > 0$ yields the reciprocal identity:

$$1 = \frac{\sum_{i=1}^n x_i}{\prod_{j=1}^n x_j} = \sum_{i=1}^n \frac{x_i}{\prod_{j=1}^n x_j} = \sum_{i=1}^n \frac{1}{\prod_{j=1, j \neq i}^n x_j}. \quad (3)$$

Step 3: Componentwise Denominator Bounds

We partition the n terms in the summation of Equation (3) into three categories:

1. **For indices $i \in \{1, 2, \dots, n-2\}$:** The denominator omits x_i and contains both x_{n-1} and x_n . Since every $x_k \geq 1$, we have:

$$\prod_{\substack{j=1 \\ j \neq i}}^n x_j = \left(\prod_{\substack{j=1 \\ j \notin \{i, n-1, n\}}}^n x_j \right) \cdot x_{n-1} x_n \geq 1 \cdot x_{n-1} x_n = x_{n-1} x_n.$$

Taking the reciprocal reverses the inequality:

$$\frac{1}{\prod_{j \neq i} x_j} \leq \frac{1}{x_{n-1} x_n} \quad \text{for each } i \in \{1, 2, \dots, n-2\}. \quad (4)$$

2. **For index $i = n-1$:** The denominator omits x_{n-1} and contains x_n :

$$\prod_{j \neq n-1} x_j = \left(\prod_{j=1}^{n-2} x_j \right) x_n \geq 1 \cdot x_n = x_n \implies \frac{1}{\prod_{j \neq n-1} x_j} \leq \frac{1}{x_n}.$$

3. **For index $i = n$:** The denominator omits x_n and contains x_{n-1} :

$$\prod_{j=1}^{n-1} x_j = \left(\prod_{j=1}^{n-2} x_j \right) x_{n-1} \geq 1 \cdot x_{n-1} = x_{n-1} \implies \frac{1}{\prod_{j=1}^{n-1} x_j} \leq \frac{1}{x_{n-1}}.$$

Step 4: Summation and Algebraic Factorization

Substituting bounds (4) through the remaining indices into Equation (3), we obtain:

$$\begin{aligned} 1 &= \sum_{i=1}^{n-2} \frac{1}{\prod_{j \neq i} x_j} + \frac{1}{\prod_{j \neq n-1} x_j} + \frac{1}{\prod_{j \neq n} x_j} \\ &\leq \sum_{i=1}^{n-2} \frac{1}{x_{n-1} x_n} + \frac{1}{x_n} + \frac{1}{x_{n-1}} \\ &= \frac{n-2}{x_{n-1} x_n} + \frac{1}{x_n} + \frac{1}{x_{n-1}}. \end{aligned}$$

Combining the right-hand side over the common denominator $x_{n-1} x_n$:

$$1 \leq \frac{(n-2) + x_{n-1} + x_n}{x_{n-1} x_n}.$$

Multiplying both sides by $x_{n-1}x_n > 0$:

$$x_{n-1}x_n \leq n - 2 + x_{n-1} + x_n.$$

Rearranging all terms involving x_{n-1} and x_n to the left-hand side:

$$x_{n-1}x_n - x_{n-1} - x_n \leq n - 2.$$

Adding 1 to both sides produces Simon's Favorite Factoring Trick form:

$$(x_{n-1} - 1)(x_n - 1) \leq n - 1. \quad (5)$$

Step 5: Elimination of the Degenerate Case $x_{n-1} = 1$

We evaluate whether $x_{n-1} = 1$ is admissible. Assume, for contradiction, that $x_{n-1} = 1$. By the non-decreasing ordering (2):

$$1 \leq x_1 \leq x_2 \leq \dots \leq x_{n-1} = 1 \implies x_1 = x_2 = \dots = x_{n-1} = 1.$$

Substituting these values directly into the original equation (1):

$$\prod_{i=1}^n x_i = 1^{n-1} \cdot x_n = x_n,$$

$$\sum_{i=1}^n x_i = (n-1) \cdot 1 + x_n = n-1 + x_n.$$

Equating both sides:

$$x_n = n-1 + x_n \iff n-1 = 0 \iff n = 1.$$

However, we are given $n \geq 2$, so $n-1 \geq 1 > 0$, rendering $x_n = n-1 + x_n$ impossible.

Hence, by contradiction:

$$x_{n-1} \geq 2 \implies x_{n-1} - 1 \geq 1.$$

Step 6: Deducing the Upper Bound

Because $x_{n-1} - 1 \geq 1$ is a positive integer, dividing inequality (5) by $(x_{n-1} - 1)$ preserves the inequality:

$$x_n - 1 \leq \frac{n-1}{x_{n-1}-1} \leq \frac{n-1}{1} = n-1.$$

Adding 1 to both sides yields:

$$x_n \leq n. \quad (6)$$

Step 7: Sharpness and Construction

To establish that n is indeed the maximum, consider the n -tuple:

$$(x_1, x_2, \dots, x_{n-2}, x_{n-1}, x_n) = (\underbrace{1, 1, \dots, 1}_{n-2 \text{ times}}, 2, n).$$

We verify Equation (1):

$$\prod_{i=1}^n x_i = 1^{n-2} \cdot 2 \cdot n = 2n,$$

$$\sum_{i=1}^n x_i = (n-2) \cdot 1 + 2 + n = n - 2 + 2 + n = 2n.$$

Both sides are identically equal to $2n$. Thus, the tuple satisfies the Diophantine condition, and the coordinate $x_n = n$ is realized. $\square$

4 Remarks

Remark 1 (The Trivial Case $n = 1$). When $n = 1$, Equation (1) reduces to $x_1 = x_1$, which holds for all $x_1 \in \mathbb{Z}^+$. Thus, the solution set is unbounded above for $n = 1$, confirming that the restriction $n \geq 2$ is necessary for the existence of a finite maximum.

Remark 2 (Uniqueness of Extremal Configuration). For equality $x_n = n$ to hold, all intermediate inequalities must hold as equalities:

1. $x_{n-1} - 1 = 1 \implies x_{n-1} = 2.$
2. $\prod_{j \neq i} x_j = x_{n-1} x_n = 2n$ for every $i \leq n-2$, which forces $x_1 = x_2 = \dots = x_{n-2} = 1.$

Thus, up to ordering, $(1, \dots, 1, 2, n)$ is the unique configuration achieving the global maximum coordinate n .